Title: EAES clinical practice guidelines 2026: Multi-society recommendations for the management of groin hernia

Running Title: EAES guidelines on the management of groin hernia

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INTRODUCTION

Inguinal hernia is the most frequently diagnosed hernia in the general population, with an overall prevalence of approximately 8%[1]. Consequently, groin hernia repair is among the most frequently performed surgical procedures worldwide. According to data from the European Commission, nearly one million groin hernia repairs were recorded across European countries in 2023[2]. Despite generally favourable outcomes after groin hernia repair, recurrence has been reported in ~2–5%, and chronic postoperative pain has been observed in around 10–20% of patients in observational cohorts, depending on the follow-up period.[3–6]. These complications contribute substantially to the global health and economic burden associated with groin hernia disease.

Despite the high volume of procedures, management of groin hernias remains variable. Over the past decade, several surgical societies—including the European Hernia Society (EHS), International Endohernia Society (IEHS), and European Association for Endoscopic Surgery (EAES)—have published guidelines for open and laparoscopic groin hernia repair, which eventually led to the international *HerniaSurge* collaboration to produce unified global recommendations[7]. The *HerniaSurge* guidelines were updated in 2023, issuing a strong recommendation in favor of laparo-endoscopic technique compared to open herniorrhaphy for primary unilateral hernias, contingent on adequate surgical expertise[8].

Recent evidence, however, highlights that, despite established guidelines supporting minimally invasive repair, adoption across Europe remains inconsistent[9]. Empirical evidence suggests similar challenges in guideline adoption in the United States. While the disparities observed at national level across European countries may be justified by national practice patterns, training standards variation, and resource availability, they may also reflect residual uncertainty in the evidence base underpinning existing recommendations. In particular, recent guidelines, including *HerniaSurge,* have drawn on a heterogeneous body of evidence, including randomized trials, systematic reviews and registry data, without conducting a de novo meta-analysis of the best available evidence. Since then, thirty new randomized trials have been published, with a total sample size of approximately 3,000 patients, comparing different laparo-

endoscopic techniques and open hernia repair, across multiple outcomes. An updated, comprehensive systematic review and meta-analysis synthesizing data from the best contemporary evidence may therefore yield more coherent comparative estimates to inform recommendations.

The current protocol outlines the initiative by EAES, the Surgical Association of Gastrointestinal and Endoscopic Surgeons (SAGES), and EHS for the development of high-quality evidence-informed recommendations to guide general surgeons and support equitable access to optimal surgical techniques across Europe.

METHODS

The protocol for the systematic review supporting this guideline adheres to the Preferred Reporting Items for Systematic reviews and Meta-Analysis Protocols (PRISMA-P)[10]. The guideline development process will follow core Grading of Recommendations Assessment, Development, and Evaluation (GRADE). Reporting of the guideline will follow Appraisal of Guidelines for Research and Evaluation (AGREE) II extension for surgical interventions (AGREE-S)[11,12]. The protocol has been registered on Open Sciences Framework (https://osf.io/zmytx).

Steering Group

The steering group will consist of three co-chairs and an additional senior methodologist. The co-chairs include a supervising methodological chair (Stavros A. Antoniou) who is an INGUIDE certified lead guideline developer and chair with expertise in abdominal wall hernia repair, a junior methodological co-chair (Alexander A. Tzanis) who is an INGUIDE certified guideline methodologist, and a surgeon with clinical and academic expertise in hernia surgery and prior participation in guideline development (Salvador Morales-Conde). The steering group will also include Gordon Guyatt, co-founder of GRADE, who will act as a senior methodologist, providing high-level methodological guidance on GRADE application, evidence synthesis, and the Evidence-to-Decision framework. Members of the steering committee will not serve as panel members and will not participate in voting. Instead, the steering committee will provide independent methodological oversight and support.

Guideline Panel

We will consider members with relevant clinical and research backgrounds, while aiming to achieve a balanced geographical representation of members from countries across Europe, as well as equal gender representation.

The guideline panel will include twelve general surgeons with expertise in groin hernia surgery: four EAES members, four EHS members, and four SAGES members. These twelve panel members will be free of significant intellectual conflicts of interest and recognized as full voting members during the synchronous consensus meeting. The panel will include three patient partners: two who have had a

previous operation for inguinal hernia, and a third patient partner who is a member of the EAES guidelines committee and a certified guideline panel member through the INGUIDE program, with participation in previous guideline panels. She will provide informal guidance to the patient partners with lived experience, with respect to the guideline panel discussion and conduct. Patient partners will be included as members of the guideline panel and will actively participate in outcome prioritization, minimal important differences, and judgements on potential benefits and harms.

All panel members, including patient partners, will receive formal preparatory training prior to participation in consensus meetings, led by the junior methodologist (AT). This will include an orientation session covering: (1) the purpose and methodology of the guideline, (2) interpretation of Summary of Findings tables, (3) the GRADE approach to rating certainty of evidence, and (4) the structure of Evidence-to-Decision discussions. To promote meaningful engagement, consensus meetings will include structured agenda items specifically inviting patient perspectives on outcome importance, acceptability of interventions, and trade-offs between benefits and harms. The Chair will actively facilitate balanced participation to ensure that patient partners contribute substantively to deliberations and final judgments.

Finally, the panel will be completed by an additional GRADE auditor who is also the leader of the AGREE collaboration for developing, reporting, and assessing the quality of clinical practice guidelines, Ivan Florez. Both Ivan Florez and Gordon Guyatt will ensure that our guideline development processes adhere to the core tenets of GRADE and will act as non-voting members. They will also review the final summary of findings tables and provide feedback. Revisions will be made according to their input, if necessary.

We will apply the International Committee of Medical Journal Editors criteria for authorship to consider any member of the panel for authorship in the peer-reviewed publication of our final guideline report[13].

Systematic Review Group

The systematic review team will be comprised of systematic reviewers, statisticians and the junior methodologist. The systematic review group will be led by the Systematic Review Lead (Konstantinos Perivoliotis) who has previous experience as systematic reviewer in guideline development projects of EAES and will coordinate and supervise all activities of the group. Four systematic reviewers will conduct literature screening, data collection and risk of bias assessment in pairs, independently. Three (Daniel Moritz Felsenreich, Michael Fadel and Stefano Granieri) have completed the INGUIDE certification for guideline panel members and are familiar with GRADE methodology, while the fourth (Jeroen van Vugt) is an experienced systematic reviewer with expertise in risk of bias assessment.

An expert statistician (Dimitris Mavridis) with experience in data analysis for over 15 clinical practice guidelines will lead the statistical analysis. This senior lead analyst will guide a junior lead analyst (Ourania Koutsiouroumpa) in performing all data analysis for the systematic review and resulting clinical practice guideline. The junior co-chair will liaise between the systematic review group and the steering group.

Conflicts of Interest

All four members of the steering group declare no personal, financial, or intellectual conflicts of interest. The steering group will collect conflicts of interest forms from the entire guideline development group including personal, intellectual, or financial conflicts. Decisions regarding management of conflicts will be documented and overseen by the steering committee. Panel members identified as having significant conflicts of interest will act as external advisors during the in-person consensus meeting and will not participate in the discussion related to the development of the final recommendations nor will they vote on the recommendations. Independently of their GRADE advisory role, they will monitor and verify that the process for managing conflicts of interest among panel members is applied appropriately, ensuring that conflicted members act only as external advisors and do not influence recommendation development or voting. The GRADE auditors will ensure that this process is handled appropriately[14]. The conflict of interest forms can be accessed in the online appendix[15].

Key Questions

This guideline topic was informed by an internal biennial survey of EAES members who vote on priority topics for future guidelines. Our steering group determined the following key questions (KQs):

1. *Should adult patients with primary groin hernia, who are candidates for elective surgical repair, be offered laparo-endoscopic or anterior open repair?*

2. *Should adult patients with primary groin hernia undergoing elective minimally invasive surgery be offered a Totally Extra-Peritoneal (TEP) or Trans-Abdominal Pre-Peritoneal (TAPP) approach?*

Subgroup analyses will be performed based on the following clinical assumptions:

- male versus female population: Shorter inguinal canal and more frequent femoral hernias in females could possibly exert more beneficial outcomes following laparoscopic repair
- body mass index (BMI) ≥35kg/m^2: The advantage of laparoscopy might be diminished over open repair in high BMI patients
- unilateral versus bilateral groin hernia: Benefit of laparoscopy may be more pronounced in bilateral hernias.
- patients diagnosed with scrotal hernia: Open repair may be more effective in scrotal hernias due to their anatomical position
- patients with prior extraperitoneal surgery (e.g. prostatectomy, placement of penile prosthesis/reservoir, artificial bladder sphincter): Dissection of the extraperitoneal space might pose risks in patients with prior lower extraperitoneal surgery.

Whenever possible, subgroup effects will be evaluated using within-trial interaction estimates, as these provide more credible evidence of effect modification. If within-trial subgroup data are unavailable, subgroup exploration may be conducted using between-trial meta-regression analyses within a pairwise meta-analysis framework.

Systematic Review

We will conduct a de novo systematic review and evidence synthesis. The Junior Co-chair will develop a systematic literature search. The Systematic Review Lead will oversee the systematic review process by leading an evidence synthesis team.

We will query online databases including PubMed, Cochrane Central Register of Controlled Trials, and Embase to identify eligible articles of interest published prior to December 7, 2025. We will use the following inclusion and exclusion criteria to select articles for inclusion for each key question:

KQ#1:

- Population: Adult population with primary unilateral or bilateral groin hernia.
- Intervention: Elective anterior open hernia repair
- Comparator: Elective laparo-endoscopic repair (TEP and TAPP)
- Study design: Randomized-controlled studies (RCTs)

KQ#2:

- Population: Adult population with primary unilateral or bilateral groin hernia.
- Intervention: TEP
- Comparator: TAPP
- Study design: Randomized-controlled studies (RCTs)

We will apply the following exclusion criteria in both KQs: we will exclude studies including 10% or more patients with the following characteristics: open preperitoneal repair approach, recurrent hernia, emergency surgery, robotic surgery, pregnant patients, and pediatric population. We will exclude studies published prior to 2015 to ensure our findings reflect contemporary evidence.

We will upload articles from all database searches to Covidence. Next, the Systematic Review Lead will pilot test the first round of screening using a sample of ten abstracts. This will be done live with the two reviewers, with iterative feedback. The same process will be repeated for the second round of screening. Similarly, the Systematic Review Lead will lead a calibration session prior to beginning data extraction using five articles. Both reviewers will perform data extraction for all included articles independently. Throughout both rounds of screening and data extraction, the Systematic Review Lead will resolve disagreements as they arise and will hold a synchronous meeting to resolve discrepancies if needed. We will extract the following variables into Covidence:

- Study Digital Object Identifier (DOI)
- Author's last name
- Year of publication
- Title
- Country
- Study design
- Inclusion criteria
- Exclusion criteria
- Age (mean ± Standard Deviation)
- American Society of Anesthesiologists (ASA)class
- BMI (kg/m^2)
- Prior lower abdominal surgery
- Prior extraperitoneal surgery
- Comorbidities
- Surgical technique (type of open repair, TEP, TAPP, type of mesh used)

- Expertise of center in advanced laparoscopic procedures
- Experience of operating surgeons in advanced laparoscopic procedures
- Outcomes (as prioritized by the panel)
    - Post-operative mortality (within 30 days)
    - 30-day major complications (Clavien-Dindo ≥ 3b)
    - Length of hospital stay
    - Quality of life
    - Recurrent groin hernia at 3 years
    - Post-operative pain (within 30 days)
    - Chronic pain at 3 months
- Follow-up for outcomes in months

Two trainee reviewers will be calibrated to perform appraisals of the risk of bias of eligible studies. Risk of bias assessments will be conducted independently for each outcome of interest or group of outcomes that share similar measurement, assessment and follow-up features. We will use the ROBUST-RCT tool for randomized controlled trials[16]. We will apply all core domains of the tool. The eight optional signalling items will not be included in the primary assessment, unless specific methodological concerns are identified that warrant their use. For each domain, reviewers will independently assess: (1) whether the relevant safeguard was implemented, and (2) if not, whether its absence introduced an important risk of bias. Domain-level judgments will reflect both considerations. Disagreements will first be resolved through discussion between the two reviewers. If consensus cannot be reached, the Systematic Review Lead will adjudicate. A synchronous meeting will be convened, if necessary, to resolve persistent discrepancies.

Outcome Prioritization and Minimal Important Differences

We searched the COMET (Core Outcome Measures in Effectiveness Trials) database[17] for a patient-centred core outcome set for groin hernia; however, at the time of the search, an eligible project was identified as being under development[18] and no finalized core outcome set was available. The Junior Co-chair has drafted a list of potential outcomes after conducting a scoping review of the literature. The

Supervising Co-chair and the Content Expert Co-chair will also suggest outcomes as appropriate. The final list of potential outcomes will be then presented to panel members, who will have the opportunity to propose additional outcomes for consideration. The panel will rate the importance of each outcome using the GRADE scale to identify important (Scores 4–6) and critical (Scores 7–9) items for inclusion[19]. In case of significant variations among panel member ratings, the steering group will hold a synchronous panel meeting to address disagreements and reach consensus. To guide this discussion, the group will follow GRADE guidance in focusing on patient-important outcomes relevant for clinicians[19].

We will distribute an online survey to panel members to define minimal important differences, according to a predefined framework[20]. We will pose questions in the following format:

"If post-operative mortality was the ONLY OUTCOME, what number of patients per 1000 would need to experience the outcome for you to consider it to be important to patients"?

- 5 per 1,000 (0.5%)
- 10 per 1,000 (1.0%)
- 50 per 1,000 (5.0%)
- 100 per 1,000 (10.0%)
- Other "[free text]"

We will review the minimal important difference thresholds derived from the online survey at the consensus meeting for approval from the panel. To decide on the outcomes to be prioritized and on the minimal important differences, we will have multiple online meetings of the guidelines development group, as needed.

Statistical analysis plan

Introduction and effect measure

We will conduct pairwise meta-analysis for each research question. For KQ#1 (open vs laparo-endoscopic repair) we will consider TEP and TAPP cases cumulatively in the laparo-endoscopic arm.

We will summarize dichotomous outcomes using odds ratios (ORs) with 95% confidence intervals (CIs). We will summarize continuous outcomes using mean differences (MDs) or standardized mean differences (SMDs) with 95% CIs when studies use different scales. We will assess between-study heterogeneity using the I^2 statistic, the chi-square statistic, the between-study variance (τ^2) and prediction intervals[21]. We will generate forest plots generated for all pairwise comparisons.

Rare events

For outcomes with low event rates (studies with zero events or probabilities events less than 0.05, e.g. 30-day mortality), we will employ a Mantel-Haenszel model using fixed effect. We will consider Peto odds ratio modeling sensitivity analysis if there are, besides low event rates, also balanced treatment groups and small treatment effects. Studies with zero events in one study arm will be included in the analysis assuming a continuity correction of 0.5, while studies with zero events in both arms will be excluded from the estimation of relative treatment effects but will be considered when assessing baseline risks and certainty of evidence. To include also double zero studies, we will employ the penalized logistic regression model of Evrenoglou et al as a sensitivity analysis[22].

We will conduct all statistical analyses using R statistical software for meta-analysis and network meta-analysis with appropriate packages (meta/netmeta/metafor) for frequentist[23]. We will present the geometry of the network through network plots, the relative effects through league tables and the ranking of interventions through rankograms.

We will conduct sensitivity analyses to evaluate the robustness of the findings, including:

- alternative statistical models for rare events (e.g., penalized logistic regression)
- exclusion of studies at high risk of bias

Results from penalized models will be compared with the primary analyses.

GRADE Certainty of Evidence

We will consider the risk of bias across studies, imprecision, inconsistency, indirectness, and publication bias to rate the certainty of evidence. To perform the certainty of evidence rating, we will use the confidence interval of effect estimates in the context of the minimal important difference thresholds, as well as the sample size of patients across comparisons to evaluate imprecision[24]. We will consider the indirectness of comparisons based on their correlation with the PICO elements according to those listed in our inclusion criteria.

We will use the MAGICapp platform to construct GRADE evidence summary tables[25].

Evidence-to-Decision (EtD) Framework

The steering group will prepare GRADE Summary of Findings tables to inform the in-person panel consensus meeting. The meeting will begin with a detailed overview of the guideline development methodology, followed by presentation of the evidence summaries. The first part of the meeting will focus on addressing questions and concerns regarding the available evidence, while the second part will focus on the evidence-to-decision framework and resolve potential disagreements by consensus[26]. The framework will consider the following factors:

- Benefits and harms of the intervention; The panel will summarize the key patient-important beneficial and harmful outcomes, and their balance, that most strongly influence the direction and strength of each recommendation.
- Certainty of the evidence; The panel will consider the overall certainty of evidence across outcomes and comparisons, highlighting outcomes with low or very low certainty and gaps in the evidence base. The relationship between certainty of evidence and conditional versus strong recommendations will be made explicit.
- Values and preferences of patients and healthcare providers; Where empirical evidence on patient values is limited, the panel will make transparent, reasoned inferences regarding the relative importance patients are likely to place on outcomes.

- Resources required, acceptability of the intervention, feasibility of implementation, and equity; The panel will consider implications for healthcare resources, surgeon training, availability of minimally invasive techniques, and equitable access to care across health systems.

A narrative summary of the EtD considerations will be presented in the main guideline for each key recommendation, following the structure above. Detailed Summary of Findings tables and additional EtD documentation will be provided in an appendix.

GRADE auditors will provide methodological guidance throughout the process for both summary of findings tables and evidence-to-decision domains, but all decisions will rest with the panel. Panel members will reach consensus on the direction and strength of recommendations through deliberation, with the opportunity to suggest revisions or additional recommendations in accordance with GRADE methodology[27]. Voting will be used in the exceptional case where unanimous agreement cannot be achieved through deliberation, with consensus defined as agreement among more than 80% of panel members. Any unresolved disagreement will be documented in the guideline report.

Target users

This guideline applies to general surgeons, other healthcare professionals, policymakers, and patients. We will develop a patient-friendly version of the guideline.

Publication and Dissemination

We will publish this clinical practice guideline *in Surgical Endoscopy and Other Interventional Techniques* as the official journal of the EAES. The steering committee will also arrange for wide dissemination via peer-reviewed scientific meetings, social media, and other channels. The steering group will consider feedback on the guideline report from a variety of sources, including social media, letters to the editor, and other feedback mechanisms. This feedback will be considered in future updates to the guideline.

Monitoring and Updates

This clinical practice guideline will be reviewed by members of the EAES one year after initial publication via an online questionnaire. Updates or extensions to the guideline will be made under the guidance of the steering committee based on the availability of new literature, and will otherwise be considered for an update five years post-publication.

Strengths and Limitations

This clinical practice guideline will be developed adhering to the highest methodological standards, including those established by GRADE. Engagement of an expert health sciences librarian will ensure that the evidence synthesis is comprehensive, rigorous, and methodologically sound. The resulting recommendations will be patient-centered and informed by the diverse perspectives of a multidisciplinary expert panel.

In terms of limitations, as with all clinical practice guidelines, interpretation of evidence inevitably involves some degree of subjectivity. To address this, an a priori–defined methodology will be employed to enhance transparency and minimize potential bias. Additionally, heterogeneity in study designs and outcomes may limit the ability to synthesize evidence conclusively.

Research ethics

All members of the guideline development group will disclose any direct (financial) or indirect (intellectual) conflicts of interest. We will manage conflicts according to the principles of the Guidelines International Network (GIN). Members with relevant direct or indirect conflicts will participate as external advisors, and they will not be involved in discussions about the direction or strength of the recommendations, the voting procedure, or the Delphi process to consolidate the recommendations.

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